A

Major project report

On

SMART INCUBATOR FOR POULTRY FARMING

Submitted in partial fulfillment of the Requirements for the award of

Degree of

BACHELOR OF TECHNOLOGY

ELECTRONICS & COMMUNICATION ENGINEERING

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Colombo Nagar, Yeshwanthapur, Jangaon-506167, Telangana

2025-2026

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CERTIFICATE

This is to certify that the report entitled **"SMART INCUBATOR FOR POULTRY FARMING"** that is being submitted by **P. UDAYASRI (22681A0443), P. SHIVASHANKAR(23685A0411), E. SAIKUMAR (22681A0421), G. VISHAL (22681A0422)** in partial fulfillment for the requirement for the award of the Degree of Bachelor of Technology in **"Electronics & Communication Engineering"** and submitted to **Christu Jyothi Institute of Technology and Science, Jangaon,** is a record of Bonafied work carried out by them under our guidance and supervision.

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- Provide highest quality learning environment to our students for their greater well-being so as to equip them with highest technical and professional ethics.
- Produce engineering graduates fully equipped to meet the ever-growing needs of industry and society

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2025-2026

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- Organize trainings and activities towards Overall personality development in time with industrial need.
- Promote innovation towards sustainable solutions with multi discipline team work with ethics.

HOD

DECLARATION

We hereby declare that the document entitled **"SMART INCUBATOR FOR POULTRY FARMING"** submitted to the **Christu Jyothi Institute of Technology and Science** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B-Tech**)** in Electronics and communication engineering is a record of an original work done by under the guidance of **Mr. B. Sandeep kumar Asst. Prof,** and this document has not been submitted to any other university for the award of any other.

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ABSTRACT

The Smart Incubator for Poultry Farming is designed exclusively for raising chicks, providing a controlled environment that ensures their healthy growth and survival. Unlike conventional incubators or brooding methods, this system focuses solely on post-hatch chick care by maintaining precise temperature and humidity levels, which are critical factors for chick development.

Equipped with temperature and humidity sensors, the incubator continuously monitors the internal environment. The NodeMCU microcontroller processes the sensor data and automatically adjusts the heating, cooling, and ventilation systems to maintain optimal conditions. Proper temperature control prevents chilling or overheating, while maintaining humidity ensures hydration and reduces stress, improving the overall health and immunity of chicks.

By automating environmental regulation, the system minimizes human intervention, reduces mortality rates, and supports uniform growth. The incubator also enhances energy efficiency by controlling the heating and cooling only when necessary. Its design makes it suitable for both small-scale and commercial poultry operations, providing a reliable, cost-effective, and scalable solution for chick rearing.

In conclusion, this smart incubator leverages precise temperature and humidity control to create an ideal, safe, and stress-free environment for chicks, ultimately improving survival rates, growth performance, and productivity in poultry farming.

CONTENTS

LIST OF FIGURES

ACRONYMS

ADC	Analog to Digital Converter
GND	Ground
ARDUINO IDE	Integrated Development Environment
DC	Direct Current
AC	Alternating Current
USB	Universal Serial Bus
DHT	dihydrotestosterone
NodeMCU	Node Microcontroller Unit
IoT	Internet of Things
AI	Artificial Intelligence
UI	User Interface
USB	Universal Serial Bus
RST	Reset
IC	Integrated Circuit
MISO	Master Input Slave Output
SDA	Serial Data Line
SCL	Seral clock Line
EN	Enable
MCU	Microcontroller Unit
ECU	Electronic Control Unit

CHAPTER 1

INTRODUCTION OF IOT

1.1 INTRODUCTION

Internet of Things (IoT) is an ideal buzzing technology to influence the Internet and communication technologies. IoT allows people and things to be connected anytime, anyplace, with anything and anyone, by using ideally in any path/network and any service. This project introduces a thought or an idea for home computerization utilizing voice acknowledgment, also the development of a prototype for controlling smart homes devices through IoT and controlling of dumb devices through IoT by the means of Wi-Fi driven chipset solution – ESP8266. This is also acknowledged by the need to give frameworks which offers help to matured and physically impaired individuals, particularly individuals who lives alone. Smart home or home automation can be said as the residential extension of building automation, it also involves the automation and controlling of lightings, ACs, ventilation and security which also includes home appliances such as dryers/washers, ovens or refrigerators/freezers which uses Wi-Fi for monitoring via remote for ease of use. Now a day's speed of the processing and communication through smart mobile devices at very affordable costs, to improve the lifestyle concept relevant to smart life, like smart T.V, Smart cities, smart phones, smart life, smart school and Internet of Things.

1.2 INTRODUCTION OF IoT TECHNOLOGY

- The Internet of Things (IoT) refers to the network of interconnected devices that communicate and share data with each other over the internet. These devices can range from everyday household items to industrial machinery.
- The term "Internet of Things" has come to describe a number of technologies and research disciplines that enable the Internet to reach out into the real world of physical objects. The Internet of Things, also called The Internet of Objects, refers to a wireless network between objects. From any time, any place connectivity for anyone, we will now have connectivity for anything

- IoT involves embedding sensors, software, and other technologies into physical objects, allowing them to collect and exchange data. This connectivity enables devices to be monitored and controlled remotely, creating a smarter and more automated world.

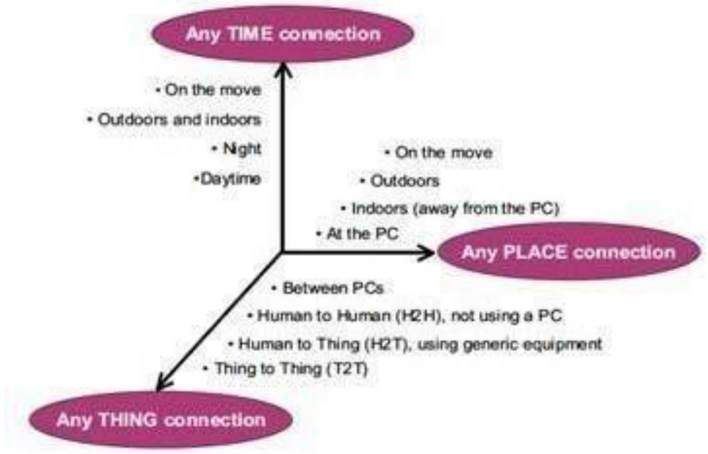


Figure 1.2: IoT Technology

1.3 The Vision

To improve human health and well-being is the ultimate goal of any economic, technological and social development. The rapid rising and aging of population is one of the macro powers that will transform the world dramatically, it has caused great pressure to food supply and healthcare systems all over the world, and the emerging technology breakthrough of the Internet-of-Things (IoT) is expected to offer promising solutions. Therefore, the application of IoT technologies for the food supply chain (FSC) (so-called Food-IoT) and in-home healthcare (IHH) (so called Health-IoT1) have been naturally highlighted in the strategic research roadmaps.

To develop practically usable technologies and architectures of IoT for these two applications is the final target of this work. The phrase "Internet of Things" (IoT) was coined at the beginning of the 21st century by the MIT Auto-ID Centre with special mention to Kevin Ashton and David L. Brock.

As a complex cyber-physical system, the IoT Integrates all kinds of sensing, identification, communication, networking, and informatics devices and systems, and seamlessly connects all the people and things upon interests, so that anybody, at any time and any place, through any device and media, can more efficiently access the information Micro Electro Mechanical Systems (MEMS), mobile internet access, cloud computing, Radio Frequency Identification (RFID), Machine-to-Machine (M2M) communication, human machine interaction (HMI), middleware, Service Oriented Architecture (SOA), Enterprise Information System (EIS), data mining, etc. With various descriptions from various viewpoints, the IoT has become the new paradigm of the evolution of information and communication technology (ICT).

1.4 Definition of Internet of things (IoT)

"Today computers and, therefore, the Internet are almost wholly dependent on human beings for information. Nearly all of the roughly 50 petabytes (a petabyte is 1,024 terabytes) of data available on the Internet were first captured and created by human being by typing, pressing a record button, taking a digital picture, or scanning a bar code.

Conventional diagrams of the Internet … leave out the most numerous and important routers of all - people. The problem is, people have limited time, attention and accuracy all of which means they are not very good at capturing data about things in the real world. And that's a big deal. We're physical, and so is our environment … You can't eat bits, burn them to stay warm or put them in your gas tank. Ideas and information are important, but things matter much more. Yet today's information technology is so dependent on data originated by people that our computers know more about ideas than things. If we had computers that knew everything there was to know about things using data they gathered without any help from us we would be able to track and count everything, and greatly reduce waste, loss and cost. We would know when things needed replacing, repairing or recalling, and whether they were fresh or past their best. The Internet of Things has the potential to change the world, just as the Internet did. Maybe even more so".

"Things are active participants in business, information and social processes where they are enabled to interact and communicate among themselves and with the environment by exchanging data and information sensed about the environment, while reacting autonomously to the real/physical world events and influencing it by running processes that trigger actions and create services with or without direct human intervention."

1.5 Key Components of IoT

- **Devices/Sensors:** These are physical objects embedded with sensors and actuators that collect and transmit data. Examples include smart thermostats, wearable fitness trackers, and industrial sensors.
- **Connectivity:** Devices connect to the internet or other networks using various technologies such as Wi-Fi, Bluetooth, Zigbee, or cellular networks.
- **Data Processing:** The data collected by IoT devices is often processed either on the device itself or sent to a cloud-based server. This data can be analyzed to gain insights, make decisions, or trigger actions.
- **User Interface:** Users interact with IoT systems through applications or dashboards, allowing them to monitor and control devices.

1.6 Applications of IoT

- **Smart Homes:** Automate and control home systems like lighting, heating, and security remotely.
- **Healthcare:** Monitor health metrics and track medication adherence with connected devices.
- **Industrial IoT:** Predict equipment failures, optimize manufacturing, and manage supply chains.
- **Agriculture:** Enhance farming with precision tools, automated irrigation, and livestock monitoring.
- **Transportation:** Track and manage vehicles, optimize routes, and support autonomous driving.
- **Smart Cities:** Improve public safety, waste management, and traffic flow with connected infrastructure.
- **Retail:** Manage inventory, personalize customer experiences, and monitor supply chains.
- **Energy Management:** Optimize energy use with smart grids and renewable energy integration.
- **Environmental Monitoring:** Track climate, pollution, and wildlife to better respond to natural events.
- **Education:** Enhance learning environments with smart classrooms and campus security systems.

1.7 Benefits of IoT

- Improved citizen's quality of life Healthcare from anywhere
- Better safety, security and productivity
- IoT can be used in every vertical for improving the efficiency
- Creates new businesses, and new and better jobs
- Economic growth
- Billions of dollars in savings and new services
- Better environment
- Saves natural resources and trees
- Helps in creating a smart, greener and sustainable planet

1.8 Characteristics for Internet of Things:

- Event driven
- Ambient intelligence
- Flexible structure
- Semantic sharing
- Complex access technology

Anyone who says that the Internet has fundamentally changed society may be right, but at the same time, the greatest transformation actually still lies ahead of us. Several new technologies are now converging in a way that means the Internet is on the brink of a substantial expansion as objects large and small get connected and assume their own web identity.

Following on from the Internet of computers, when our servers and personal computers were connected to a global network, and the Internet of mobile telephones, when it was the turn of telephones and other mobile units, the next phase of development is the Internet of things, when more or less anything will be connected and managed in the virtual world.

Smart connectivity with existing networks and context-aware computation using network resources is an indispensable part of IoT. With the growing presence of Wi-Fi and 4G-LTE wireless Internet access, the evolution towards ubiquitous information and communication networks is already evident. However, for the Internet of Things vision to successfully emerge, the computing paradigm will need to go beyond traditional mobile computing scenarios that use smart phones and portables, and evolve into connecting everyday existing objects and embedding intelligence into our environment. For technology to disappear from the consciousness of the user, the Internet of Things demands: a shared understanding of the situation of its users and their appliances, software architectures and pervasive communication networks to process and convey the contextual information to where it is relevant, and the analytics tools in the Internet of Things that aim for autonomous and smart behavior. With these three fundamental grounds in place, smart connectivity and context-aware computation can be accomplished.

CHAPTER 2

LITERATURE SURVEY

Automated Incubation Systems Li et al. (2020) developed an IoT-enabled incubator that monitors and controls temperature and humidity in real time, increasing hatch rates by about 10%.

Sensor-Based Monitoring Kumar & Singh (2019) implemented incubators with DHT22 sensors for temperature and humidity control, integrated with microcontrollers to automate operations and reduce human errors.

Mobile and Cloud Integration Chen et al. (2021) created a mobile-based incubator control system that allowed farmers to monitor parameters remotely, enhancing convenience and control.

Predictive Analytics in Incubation Zhang & Wang (2022) applied AI models to predict hatching outcomes based on sensor data, improving incubation accuracy and efficiency.

IoT and Sensor Integration: Smart incubators commonly use microcontrollers (such as ESP32) and sensors (DHT22/DHT11) for real-time monitoring and control of temperature and humidity, with automation replacing manual adjustments.

PID & AI-Based Control: Advanced control techniques, such as PID controllers (tuned via Ziegler–Nichols) and fuzzy logic AI, dynamically regulate temperature and humidity, improving hatch rates and maintaining stable incubation environments.

Remote Monitoring & Automation: Mobile applications and dashboards (often built using platforms like Blynk) allow farmers to monitor, adjust, and receive notifications about hatch conditions from anywhere, minimizing manual labor and reducing human errors.

Systems may include automated food dispensing and egg-turning mechanisms to ensure uniform heating and healthy chick growth during and after hatching.

Many researches and revolutions have resulted in developing automated incubation devices. The majority of research emphasized keeping a temperature range of 37.5

degrees Celsius to 38 degrees Celsius for optimum embryonic development. In such kind of setups, sensor-based solutions like DHT22 were applied for humidity and DS18B20 for temperature to ensure proper incubation. For poultry, it brought so much success to hatchability and reduced chick mortality through improved control of the environment during incubation.

Modern developments of IoTs made an improvement to incubators by enabling real time monitoring and remote control via mobile applications. Studies have shown anchoring water pumps with temperature sensors to the incubator maintains humidity levels within the required limits, which prevents egg dehydration. Timer-based food dispensers have also been applied in poultry farming to dispense food according to the time for feeding-growing chicks. This combination of automating the temperature, humidity, and feeding would greatly increase efficiency and productivity on poultry incubation on the farm level.

- **IoT-based monitoring:** Central to these systems is the use of IoT technology to provide remote, real-time monitoring of the brooder environment. Microcontrollers like the Raspberry Pi Pico and ATmega2560 are used to process data from various sensors.
- **Multi-sensor integration:** Smart brooders rely on multiple sensors to accurately capture environmental conditions. Common sensors include:
- **Temperature sensors (e.g., DHT22):** To monitor and regulate the temperature, which is critical for chick survival, especially in the first few weeks.
- **Humidity sensors:** To control moisture levels, as high humidity can increase the risk of microbial growth and disease.
- **Gas sensors (e.g., MQ-135):** To detect harmful gases like carbon dioxide and ammonia that can negatively impact chick health.
- **Infrared (IR) sensors:** Used for motion detection, sometimes to detect predator

CHAPTER 3

OVER VIEW OF THE PROJECT

3.1 INTRODUCTION

A Smart Incubator for Poultry Farming is an automated system designed to create and maintain the ideal environmental conditions required for hatching poultry eggs efficiently. The main goal of this project is to improve hatching success rates while reducing manual effort through the use of sensors, microcontrollers, and control mechanisms.

In traditional incubation, farmers manually monitor and adjust parameters such as temperature, humidity, and egg turning, which can lead to errors and inconsistent results. A smart incubator eliminates these problems by automatically controlling these parameters using embedded systems and sensor technology.

The system typically uses sensors to measure temperature and humidity inside the incubator. These readings are processed by a microcontroller (such as Arduino or NodeMCU), which activates devices like heaters, fans, and humidifiers to maintain optimal conditions. An automatic egg-turning mechanism ensures that eggs are rotated periodically to prevent the embryo from sticking to the shell. Some systems also include IoT features for remote monitoring and control via a mobile application or web interface.

The ideal incubation conditions for chicken eggs are around 37.5°C temperature and 55–65% humidity, which must be maintained throughout the incubation period (about 21 days). The smart incubator ensures these parameters remain stable, resulting in higher hatchability rates.

This project promotes automation, energy efficiency, and precision in poultry farming. It is especially beneficial for small and medium-scale farmers as it reduces labor costs, minimizes human error, and enhances productivity. By using locally available materials and affordable electronics, the smart incubator can be made cost-effective and sustainable, supporting modern agricultural practices and rural development.

This project contributes significantly to modernizing poultry farming by integrating technology with agriculture. It not only improves the quality and quantity of chick production but also supports sustainable and intelligent farming practices. Such a system can be beneficial for small-scale farmers, educational institutions, and agricultural research centers aiming to achieve efficient and reliable poultry production.

The successful artificial incubation of eggs is governed by four key principles that the smart incubator is designed to automate and control:

- **Temperature regulation:** The embryo can only withstand small temperature fluctuations during development. A smart incubator uses sensors and a microcontroller to maintain a precise and stable temperature, typically around 37.5°C (99.5°F) for chicken eggs.
- **Humidity control:** Maintaining optimal humidity levels prevents excessive moisture loss from the eggs. Humidity varies throughout the incubation cycle and is often higher during the final days before hatching. The smart incubator automates this process using a mist maker or water reservoir system.

Actuation: The microcontroller uses relays to act as switches for the output components.

- **Heating bulb:** The bulb is used as a heat source. The relay turns the bulb on or off based on the temperature reading.
- **Fan:** The fan circulates the warm, moist air to ensure even temperature and humidity distribution throughout the incubator.
- **Monitoring and Feedback:** An LCD screen displays the current temperature and humidity readings. For more advanced projects using Internet of Things (IoT) technology, the data can be sent to a server and accessed via a smartphone app for remote monitoring.
- **Light Sensor:** A simple light-dependent resistor (LDR) can be used to detect light levels.
- **Microcontroller:** Processes the data from the LDR.

- **LCD or LED:** A simple light can indicate the status of the incubator (e.g., green for optimal, red for alert).

  - **Continuous monitoring:** The temperature and humidity sensors provide a constant stream of environmental data.

  - **Automated adjustments:** The microcontroller acts as the brain, automatically adjusting the bulb (heater) and fans to respond to changes in the environment.

  - **Error handling:** An alarm system (visual or audible) can be integrated to notify the user if environmental conditions deviate significantly from the set parameters.

3.2 EXISTING SYSTEM

In the existing system of poultry incubation, most of the processes involved in hatching eggs are carried out manually without the support of automated or smart control technologies. The maintenance of suitable environmental conditions such as temperature, humidity, and ventilation depends entirely on the experience and attention of the operator. Farmers usually use basic instruments like thermometers and hygrometers to check temperature and humidity levels inside the incubator. Any adjustment to these parameters is done manually, which requires constant observation and effort.

The cheeks are turned by hand multiple times a day to ensure that the embryo develops uniformly. However, failure to turn the eggs at the right intervals can lead to poor hatching rates or embryo death. Similarly, the water tray used to maintain humidity must be refilled regularly, as there is no automatic mechanism to regulate it. There is no feedback or alert system to indicate abnormal conditions, which makes it difficult to maintain consistent incubation parameters.

Because the process depends heavily on human effort, the chances of error are high. Slight changes in environmental conditions can adversely affect the development of the eggs, leading to low hatching efficiency. Moreover, continuous monitoring increases the workload of the farmer and makes the process time-consuming.

Thus, the existing system is not efficient for large-scale poultry production. It lacks automation, accuracy, and reliability. The dependence on manual control not only reduces productivity but also increases operational costs. Therefore, there is a need for a smart automated system that can monitor and control temperature, humidity, and egg rotation automatically to improve the efficiency and success rate of poultry incubation.

- **Limitations:**
  - **Manual operation:**
    All processes such as temperature control, humidity maintenance, and egg turning are done manually, which requires constant human effort.
  - **Lack of Automation:**
    There are no sensors or microcontrollers to automatically regulate environmental conditions inside the incubator.
  - **Inaccurate Control:**
    Manual adjustments of temperature and humidity are not precise, leading to unstable incubation conditions.

3.3 PROPOSED SYSTEM

The proposed system is designed to automate and improve the traditional poultry incubation process by integrating smart technologies, microcontrollers, and IoT-based monitoring. The main objective of this system is to provide a controlled and stable environment for egg hatching with minimal human intervention. It ensures that the key incubation parameters such as temperature, humidity, and egg rotation are maintained automatically and precisely throughout the incubation period.

In this system, a NodeMCU (ESP8266 or ESP32) microcontroller acts as the central control unit. It receives input signals from different sensors — such as the DHT11 or DHT22 for temperature and humidity sensing — and processes the data to maintain the required conditions. The microcontroller compares the sensed values with the predefined set points. If there is any deviation, it automatically triggers actuators such as heaters, cooling fans, or humidifiers to restore the ideal environment.

The proposed system also includes an LCD or OLED display to show real-time readings of temperature, humidity, and system status. Moreover, through IoT integration, the data can be accessed remotely via a mobile application or web interface, allowing the user to monitor and control the incubator from anywhere using Wi-Fi. This provides great convenience and ensures that any abnormal condition can be addressed immediately through alerts or notifications.

Additionally, the system can store environmental data for analysis, helping farmers understand the incubation process better and make informed decisions for future batches. The automation improves consistency, efficiency, and reliability of hatching operations while reducing human effort and errors.

Fig 3.3.1: Poultry farming

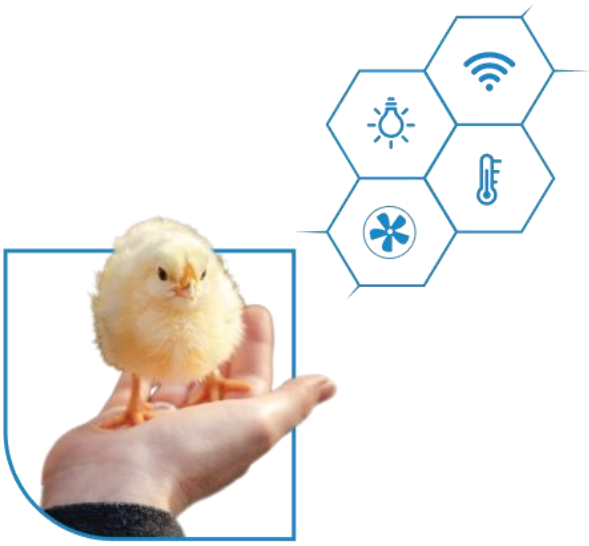


Fig 3.3.2: Farm Automation

3.3.3 BLOCK DIAGRAM OF PROPOSED SYSTEM

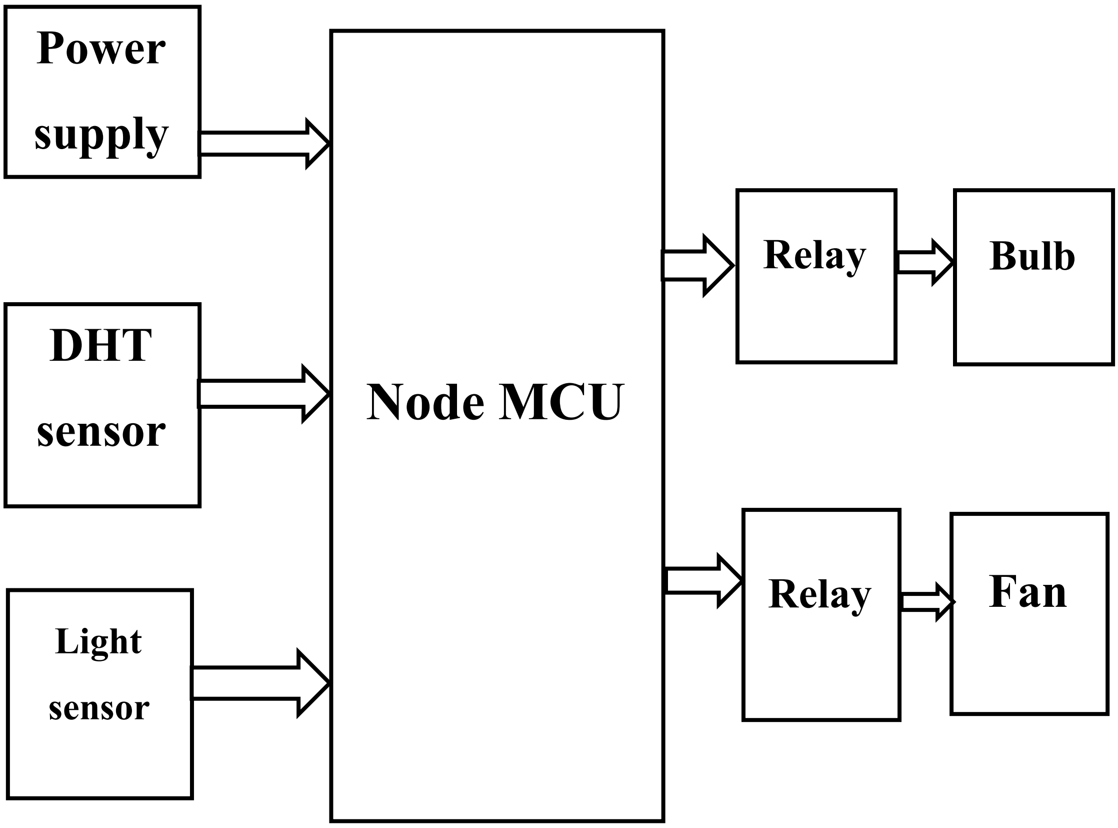


Fig 3.3.3: Block diagram of Smart Incubator for Poultry Farming

Hardware Components

- NodeMCU
- DHT11 Sensor
- Light Sensor
- Relay2
- Bulb
- Fan

Software Components

- Arduino IDE
- Blynk

3.4 WORKING PRICIPLE OF PROPOSED SYSTEM

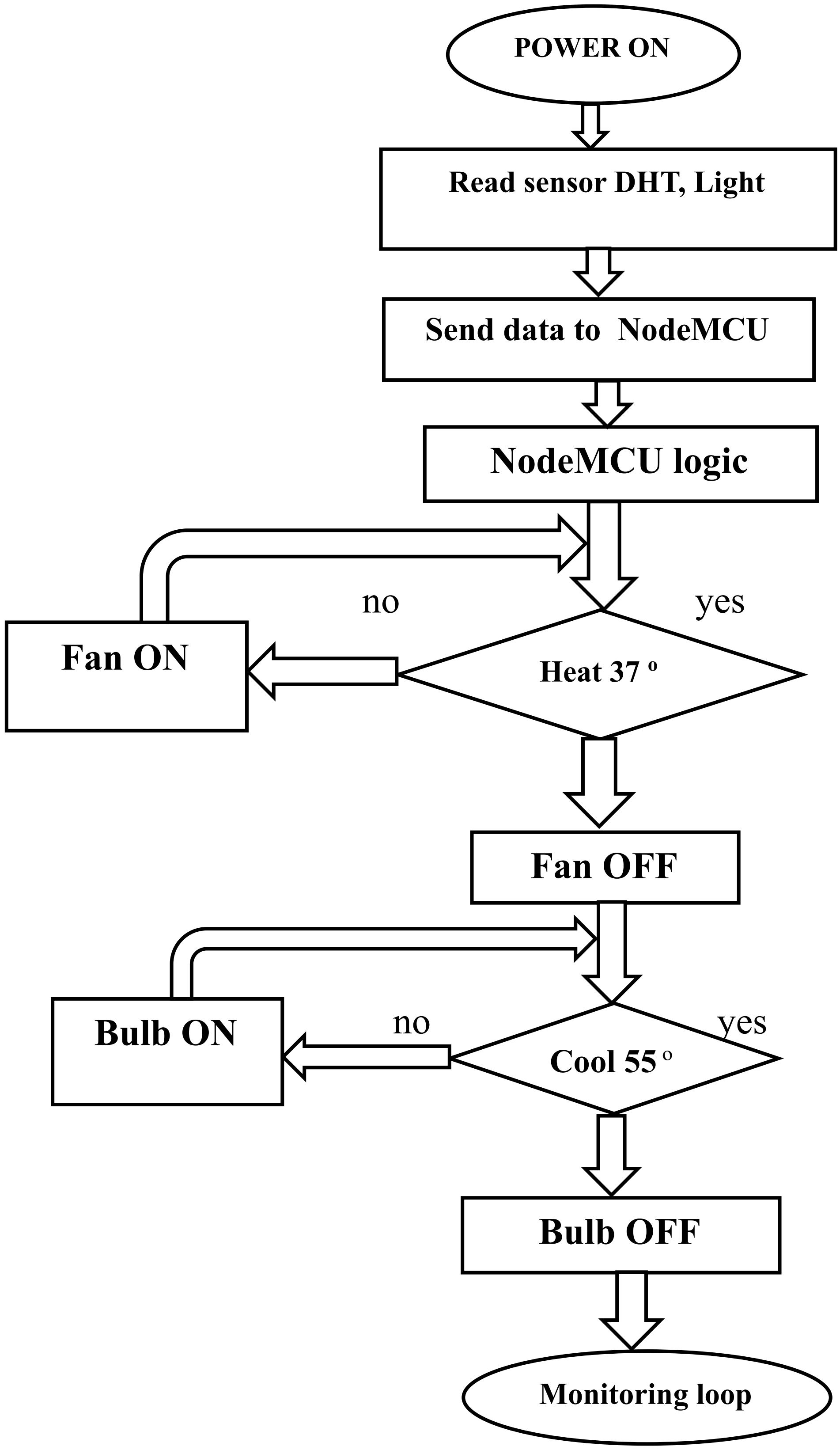


Fig 3.4: working principle of Smart Incubation for Poultry Farming

Explanation:

The flowchart explains the operational logic for a smart incubator designed for poultry farming. The system uses a NodeMCU microcontroller to automate the process of monitoring and controlling the environment inside the incubator.

The process flow is as follows:

- **Start:** The system is powered on.
- **Sensor Reading:** The system reads data from multiple sensors, including a DHT sensor (for temperature and humidity), a light sensor, and a water level sensor.
- **Data Processing:** The collected data is sent to the NodeMCU, which acts as the control unit for the system.
- **Temperature Control:** The NodeMCU logic checks the temperature.
    - If the temperature is greater than 37°C, the "Fan ON" action is triggered to cool the incubator. Once the temperature drops, the fan is turned "OFF."
    - If the temperature is less than 37°C, the system might activate a heat source (implied by the "Heat 37d" diamond), often a bulb, to warm the incubator to the desired temperature.
- **Humidity Control:** The NodeMCU logic also checks the humidity, which is set to a specific level (55d in the diagram).
    - If the humidity is too high, the system may turn a fan "ON" to circulate air and reduce humidity.
    - If the humidity is too low, a bulb might be turned "ON" to increase the temperature and, consequently, the humidity. The diagram shows a "Bulb ON" action followed by "Bulb OFF" after the humidity level is met.
- **Water Level Monitoring:** The water level sensor is checked to ensure there is enough water to maintain humidity levels.

- **Monitoring Loop:** The entire process is a continuous "Monitoring loop," where the system constantly reads sensor data and adjusts the fan and bulb to maintain the optimal conditions for the eggs.

A smart incubator for poultry farming uses sensors to monitor temperature and humidity, which a microcontroller uses to automatically control heating elements, fans, and to maintain optimal hatching conditions. These systems are often connected to the internet (IoT) to send data to a user's smartphone or PC, allowing for remote monitoring, alerts, and control of the incubator from anywhere in the world.

- **Remote access:**
    - Data is sent to a cloud platform or data store.
    - Users can access this data through a web browser or a dedicated app on their smartphone or computer.
    - Users receive automated alerts for any deviations from the ideal environment.
    - Some systems allow users to remotely adjust the settings, like temperature or turning frequency.

When using chicks for poultry farming, a smart brooder is the appropriate tool for intelligent management. A smart brooder works by automatically controlling and monitoring the chicks' environment to ensure they grow in optimal conditions, reducing chick mortality and increasing efficiency for the farmer.

- **Maximizes hatch rates:** By precisely and consistently controlling all environmental variables, smart incubators drastically increase the percentage of viable chicks that hatch.
- **Reduces labor:** Automation reduces the need for constant, manual monitoring and human intervention, saving farmers significant time and effort.
- **Ensures consistency:** Unlike natural brooding, which can be inconsistent, the controlled environment of a smart incubator provides stable and reliable conditions for every batch of eggs.

CHAPERT 4
HARDWARE COMPONENTS

4.1 POWER SUPPLY

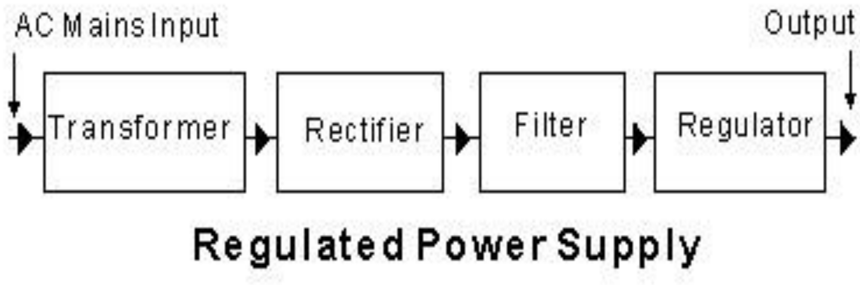


Fig 4.1: power supply

Transformer

- A transformer is an electrical device that transfers electrical energy from one circuit to another through electromagnetic induction.
- It's used to step up or step down the input voltage to the required level.

Rectifier

- A rectifier is an electrical device that converts AC (Alternating Current) voltage to DC (Direct Current) voltage.
- The rectifier uses diodes to convert the AC voltage to DC voltage.

Filter

- A filter is an electrical device that removes unwanted components from a signal.
- In a power supply, the filter is used to remove ripples and noise from the DC voltage.

Regulator

Regulators are available in different types, including linear regulators, switching regulators, and voltage references.

4.2 NodeMCU

A NodeMCU is an open-source platform that includes firmware (originally for LUA), a software development environment, and an open-source hardware development board for the ESP8266 Wi-Fi System-on-a-Chip (SoC). It is popular for Internet of Things (IoT) projects because the ESP8266 provides Wi-Fi connectivity, acting as a low-cost, self-contained computer. The development boards are designed for easy prototyping, featuring Analog (A0), Digital (D0-D8), and other pins for connecting external components.

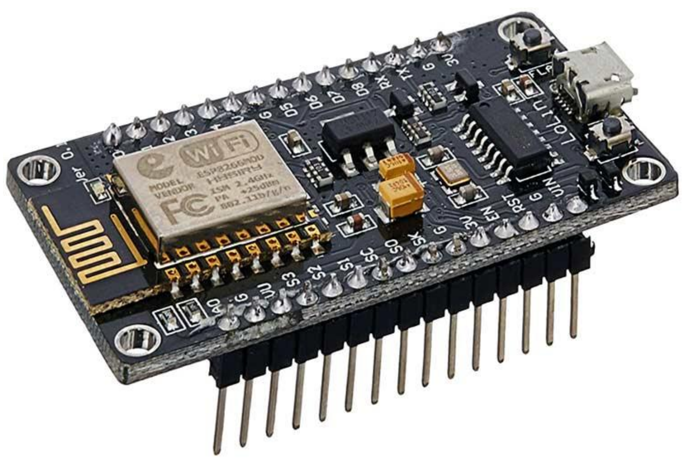


Fig 4.2: NodeMCU

Applications:

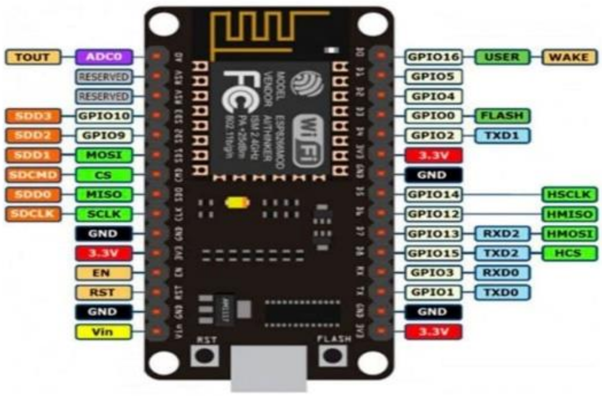
NodeMCU is widely used in various IoT and automation projects thanks to its affordability, ease of integration, and Wi-Fi features. Some common applications include:

- Home automation systems (controlling lights, appliances, HVAC).
- Energy monitoring and smart metering solutions.
- Smart gardens (soil moisture sensing, automated watering).
- Remote environmental monitoring (temperature, humidity sensors).
- Security and surveillance devices, including motion sensors and smart cameras.

Specifications:

- The Wi-Fi module is compatible with the 802.11 b/g/n standard at 2.4 GHz, has an integrated TCP/IP stack, 19.5 dBm output power, data interface (UART / HSPI / I2C / I2S / Ir Remote Control GPIO / PWM) and PCB antenna. It also has a micro USB connector and reset button.

4.2.1 PIN DESCRIPTION OF NODEMCU



Vin – input voltage 5V to the NodeMCU

3V3 – 3.3V output to power sensors

GND – ground pin

D0 (GPIO16) – digital pin, used for wake-up, no PWM or I2C

D1 (GPIO5) – used for I2C SCL (clock line)

D2 (GPIO4) – used for I2C SDA (data line)

D3 (GPIO0) – must be HIGH during boot, used for flashing

D4 (GPIO2) – connected to onboard LED (active LOW)

D5 (GPIO14) – can be used as PWM or SPI SCK

D6 (GPIO12) – can be used as PWM or SPI MISO

D7 (GPIO13) – can be used as PWM or SPI MOSI

D8 (GPIO15) – must be LOW during boot, used as SPI SS

A0 (ADC0) – analog input pin, reads 0–1V range

RX (GPIO3) – receives serial data

TX (GPIO1) – transmits serial data

SCL → D1 (GPIO5) – I2C clock line

SDA → D2 (GPIO4) – I2C data line

SCK → D5 (GPIO14) – SPI clock

MISO → D6 (GPIO12) – SPI data input

MOSI → D7 (GPIO13) – SPI data output

SS → D8 (GPIO15) – SPI slave select

EN – enable pin, must be HIGH for module to run

RST – reset pin, used to restart the microcontroller

4.3 DHT11 SENSOR

The DHT11 is a low-cost digital sensor that measures both relative humidity and temperature, using a capacitive humidity sensor and a thermistor to detect environmental changes. It outputs digital signals, making it easy to interface with microcontrollers, and is used in applications like weather stations, smart homes, and climate control.

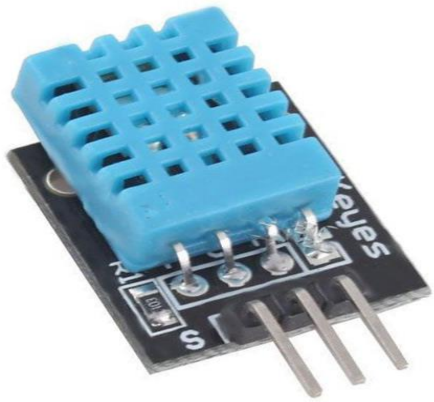


Fig 4.3: DHT11 Sensor

- **Humidity Measurement:**

A capacitive humidity sensor uses a moisture-holding substrate between two electrodes; changes in humidity cause a corresponding change in the substrate's capacitance.

- **Temperature Measurement:**

A negative temperature coefficient (NTC) thermistor's resistance decreases as the temperature increases.

- **Digital Signal Processing:**

The on-board 8-bit microprocessor measures the changes in capacitance and resistance, processes these values, and converts them into a digital serial data stream.

Applications:

- Weather Monitoring Systems
- Smart Incubators

- Greenhouse Automation
- Home Automation Systems
- Agriculture and Farming
- HVAC (Heating, Ventilation, and Air Conditioning)
- Industrial Environmental Monitoring
- Data Logging Systems

Specifications:

- Type: Temperature and Humidity Sensor
- Temperature Range: 0°C to 50°C
- Temperature Accuracy: ±2°C
- Humidity Range: 20% to 90% RH (Relative Humidity)
- Humidity Accuracy: ±5% RH
- Operating Voltage: 3.3V to 5.5V DC
- Operating Current: 0.3 mA (measuring), 60 µA (standby)
- Sampling Rate: 1 Hz (one reading per second)

4.4 LIGHT SENSOR:

A light sensor, or photoelectric sensor, detects and measures light, converting it into an electrical signal that can control devices or provide information about light intensity, color, or its presence.

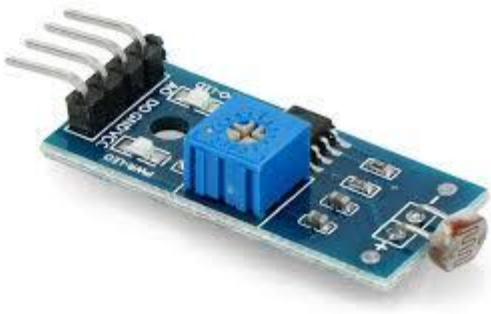


Fig 4.4: Light Sensor

Common types include photodiodes, photoresistors, and phototransistors, and they are widely used in smartphones to adjust screen brightness, in cameras, and in security systems.

Automatic Brightness Adjustment is used in phones and tablets to sense the ambient light and adjust the screen's brightness for optimal viewing and power saving. Cameras adjust the aperture of a camera to control the amount of light that enters the lens. Security Systems detect the presence or absence of light to identify potential intrusions.

Industrial Automation is used in various automated systems where the presence or absence of an object needs to be detected via a light beam.

Scientific Experiments is used in educational settings to demonstrate principles of light, such as the inverse square law.

Specifications:

- **Type:** LDR (Light Dependent Resistor) / Photodiode / Phototransistor
- **Material:** Cadmium Sulfide (CdS) or similar light-sensitive semiconductor
- **Operating Voltage:** Typically 3V to 5V DC
- **Resistance in Light:** Around 1 kΩ to 10 kΩ (low resistance when light is bright)
- **Resistance in Dark:** Around 0.5 MΩ to 10 MΩ (high resistance in darkness)
- **Spectral Response Range:** 400 nm to 700 nm (visible light region)
- **Peak Sensitivity Wavelength:** Around 550 nm (green light)
- **Operating Temperature Range:** –30°C to +70°C

Applications:

- Automatic Street Lighting
- Smart Home Lighting Systems
- Solar Tracking Systems
- Display Brightness Control
- Security and Alarm Systems

4.5 RELAY:

A relay is an electromechanical switch used to control a high-voltage or high-current circuit using a low-voltage signal.

- It works on the principle of electromagnetic induction - when current flows through a coil, it creates a magnetic field that moves a switch (contacts) to open or close another circuit. It works like a bridge between low power circuits (microcontrollers) and high power circuits (appliances).
- When a small current flows through the relay's coil, it creates a magnetic field.
- This magnetic field pulls a switch, allowing a larger current to flow through another circuit.

- It helps control high-power devices safely without directly connecting them to sensitive components.

Fig 4.5: Relay2

Specifications:

- **Number of Pins:** 3 (COM, NO, and Coil)
- **Operating Voltage:** 5V / 6V / 12V / 24V DC or AC (depending on relay type)
- **Operating Current:** Typically 30–100 mA for coil activation
- **Contact Current Rating:** 5A to 10A (for switching loads)
- **Contact Voltage Rating:** Up to 250V AC or 30V DC
- **Contact Type:** SPST (Single Pole Single Throw) — one circuit controlled by one switch
- **Coil Resistance:** 70–400 Ω (depends on coil voltage rating)

Applications:

- Home Automation Systems :Controls lights, fans, and appliances remotely.
- Industrial Control Systems: Used to control motors, heaters, and heavy machinery.
- Automotive Applications: Found in car headlights, horn circuits, and fuel pumps.
- Smart Incubators Controls heating elements and fans based on temperature/humidity sensors.

4.6 BULB:

The main function of an electric bulb is to convert electrical energy into light energy for illumination. It does this by passing an electric current through a filament or gas, which heats up to a high temperature and glows, emitting visible light.

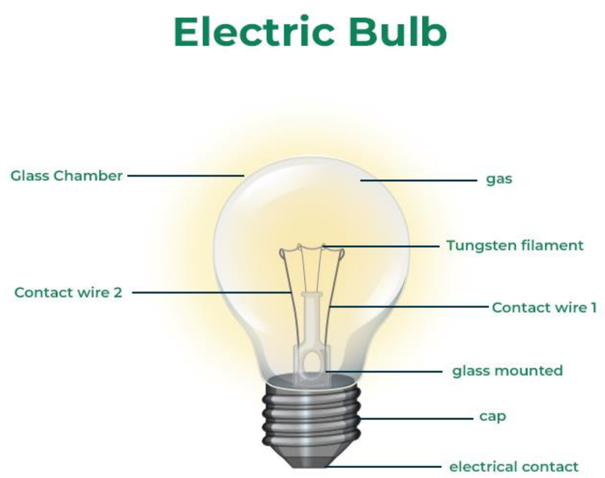


Fig 4.6: Bulb

Applications:

- **Smart light control:** Apps like Bulb - Smart Light Control allow users to control smart lights from brands like Philips Hue, LIFX, and Gove, often with options to sync colors and music.
- Household lighting (living rooms, bedrooms)
- Decorative lighting (chandeliers, lamps)
- Heat lamps (for food warming or reptile terrariums)
- Signal lights in appliances (oven lights, refrigerators)

Specifications:

- **Voltage:** The operating voltage the bulb requires.
  - **Unit:** Volts (V)
  - **Common Values:**
    - Household: 110–120V (US), 220–240V (India)
    - Low voltage: 12V or 24V (LED strips, specialty bulbs)

Base Type: The socket type that fits the bulb.

Examples:

- **Edison Screw (E27, E14)**
- **Bayonet Cap (B22)**

4.7 FAN:

Fig 4.7: Fan

At its most basic, a fan consists of rotating blades driven by a motor to move air or other gases. The device works by drawing air in from one side (the suction side) and expelling it out the other (the pressure side). Different types of fans, such as axial and centrifugal fans, are categorized based on their design and the direction of the airflow they create.

Applications:

They ensure efficient circulation of conditioned air and regulate temperature, humidity, and air quality within commercial and industrial setups. Drying and Cooling: Industrial process fans are widely used in drying and cooling applications all over.

Specifications:

- Voltage: The electrical power the fan requires to operate.
- Wattage/Power Consumption: The amount of power the fan uses.
- Frequency: For AC fans, the frequency (e.g., 50 Hz) influences speed.
- Speed: Measured in RPM (revolutions per minute) or CMM (cubic meters per minute), indicating how fast the blades spin and how much air is moved.
- Sweep Size (Ceiling Fans): The diameter of the area the fan blades cover, determining its effectiveness in a given room size.

CHAPTER 5

SOFTWARE COMPONENTS

5.1 ARDUINO IDE

Arduino is an open-source electronics platform that combines easy-to-use hardware and software to facilitate the creation of interactive electronic projects. It features a range of microcontroller boards, like the Arduino Uno, which are programmed through the Arduino IDE using a simplified version of C/C++. This userfriendly approach, along with extensive libraries and community support, makes Arduino ideal for prototyping, educational purposes, and hobbyist projects, enabling users to interface with sensors, actuators, and other components to build various applications.

- **Cross-platform:** Works on multiple operating systems (Windows, macOS, Linux).
- **Free and Open-source:** Free to use, modify, and distribute.
- **User-friendly:** Easy to use, even for beginners.
- **Syntax Highlighting:** Color-codes code for easier reading.
- **Auto-completion:** Suggests code completions as you type.
- **Code Folding:** Hides and shows sections of code.

The key features are:

- Arduino board are able to read analog or digital input signals from different sensors and turn it into an output such as activating a motor, turning LED on/off, connect to the cloud and many other actions.
- Can control your board functions by sending a set of instructions to the microcontroller on the board via Arduino IDE (referred to as uploading software).
- Unlike most previous programmable circuit board, Arduino does not need an extra piece of hardware (called a programmable) in order to load a new code onto the board. You can simply use a USB cable.
- Additionally, the Arduino IDE uses a simplified version of C++, making it easier to learn to program.
- Finally, Arduino provides a standard from factors that breaks the functions of the microcontroller into a more accessible package.

After learning about the main parts of the Arduino UNO board. We are ready to learn how to set up the Arduino IDE. Once we learn the, we will ready to upload our program on the Arduino board.

5.1.1 ARDUINO DATA TYPES:

Data types in C refers to an extensive system used for declaring variable or functions of different types. The type of a variable determines how much ace it occupies in the storage and how the bit pattern stored is interpreted.

The following table provides all the data types that you will use during Arduino programming.

Void:

The void keyword is used only in function declaration. It indicates that the functions is expected to return no information to the functions from which it was called.

Example:

```
Void Loop ()
{
// rest of t code
}
```

Boolean: A Boolean holds one of two values, true or false. Each Boolean variable occupies one byte of memory.

Chart: A data type that takes up one byte of memory that stores a character value.

Character literals are written in single quotes like this: ‘A’ and for multiple characters, strings use double quotes: “ABC”.

However, characters are stored as numbers. You can see the specific encoding in the ASCII chart. This means that it is possible to do arithmetic operations on characters, in which the ASCII value of the character is used. For example, ‘A’ +1 has the value 66, since the ASCII value of the capital of the capital letter a is 65.

Unsigned char is an unsigned data type that occupies one byte of memory. The unsigned char data type encodes numbers from 0 to 255. **Byte:** A byte stores an 8-bit unsigned number, from 0 to 255.

Int: Integers are the primary data-type for numbers storage. int stores a 16-bit (2-byte) value. This yields a range of -32,768 to 32,767 (minimum value of -2^15 and a maximum value of (2^15)-).

The Int size varies from board to board. On the Arduino due, for example, an int stores a 32-bit (4-byte) value.

Unsigned int: Unsigned int (unsigned integers) are the same as int in the way that they store a 2byte value. Instead of storing negative numbers, however, they only store positive values, yielding a useful range of 0 to 65,535 (2^16). The due stores a 4byte (32-bit) value, ranging from 0 to 4,294.967.295 (2^32-).

Word: On the Uno and other A TMEGA based boards, a word stores a 16-bit unsigned number. On the due and zero, it stores a 32-bit unsigned number.

Long: Long variables are extended size variable for number storage, and store 32 bit (4 bytes), from 2,147,483,648 to 2,147,483,647.

Unsigned long: Unsigned long variables are extended size variables for number storage and store 32 bits (4 bytes). Unlike standard longs, unsigned longs will not store negative numbers, making their range from 0 to 4,294,967,295 (2^32 - 1).

Short: A short is a 16-bit data-type. On all Arduinos (AT Mega and ARM based), a short stores a 16-bit (2-byte) value. This yields a range of -32,768 to 32,767 (minimum value of -2^15 and a maximum value of (2^15) - 1).

Float: Data type for floating-point number is a number that has a decimal point. Floating-point numbers are often used to approximate the analog and continuous values because they have greater resolution than integers. Floating-point numbers can be as large as 3.4028235E+38 and as low as 3.4028235E+38. They are stored as 32 bits (4 bytes) of information.

Double: On the Uno and other ATMEGA based boards, Double precision floatingpoint number occupies four bytes. That is, the double implementation is exactly the same as the float, with no gain in precision. On the Arduino Due, doubles have 8-byte (64 bit) precision.

In this section, we will learn in easy steps, how to set up the Arduino IDE on our computer and prepare the board to receive the program via USB cable.

Step 1: First you must have your Arduino board (you can choose your favorite board) and a USB cable. In case you use Arduino UNO, Arduino duemilanove, Nano, Arduino Mega2560, or Decimal, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image.

Fig 5.1: USB cable

Step 2: Download Arduino IDE Software.

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.

Step 3: Power up your board.

The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an Arduino Decimal, you have to make sure that the board is configured to draw power from the USB connection. The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port. Connect the Arduino board to your computer using the USB cable

Step 4: Launch Arduino IDE.

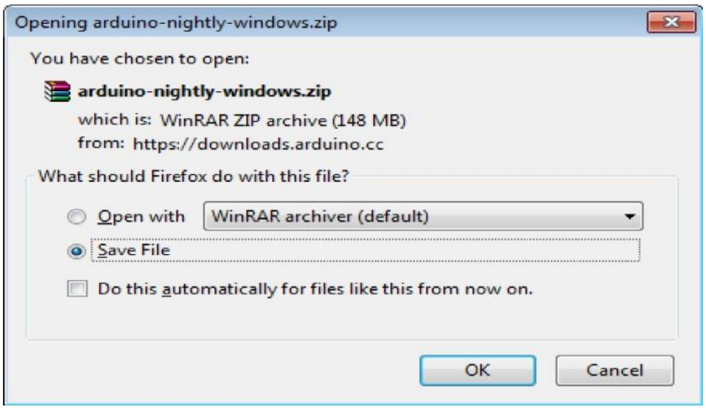
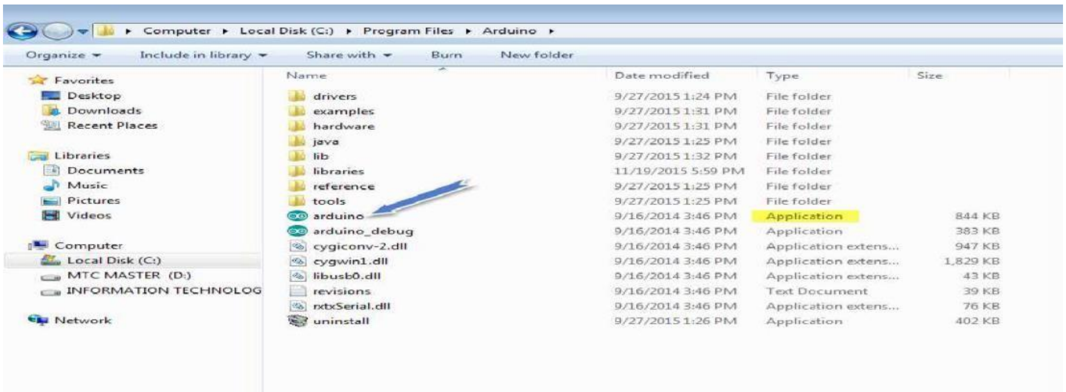


Fig 5.2: Launch Arduino IDE

After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double click the icon to start the IDE.

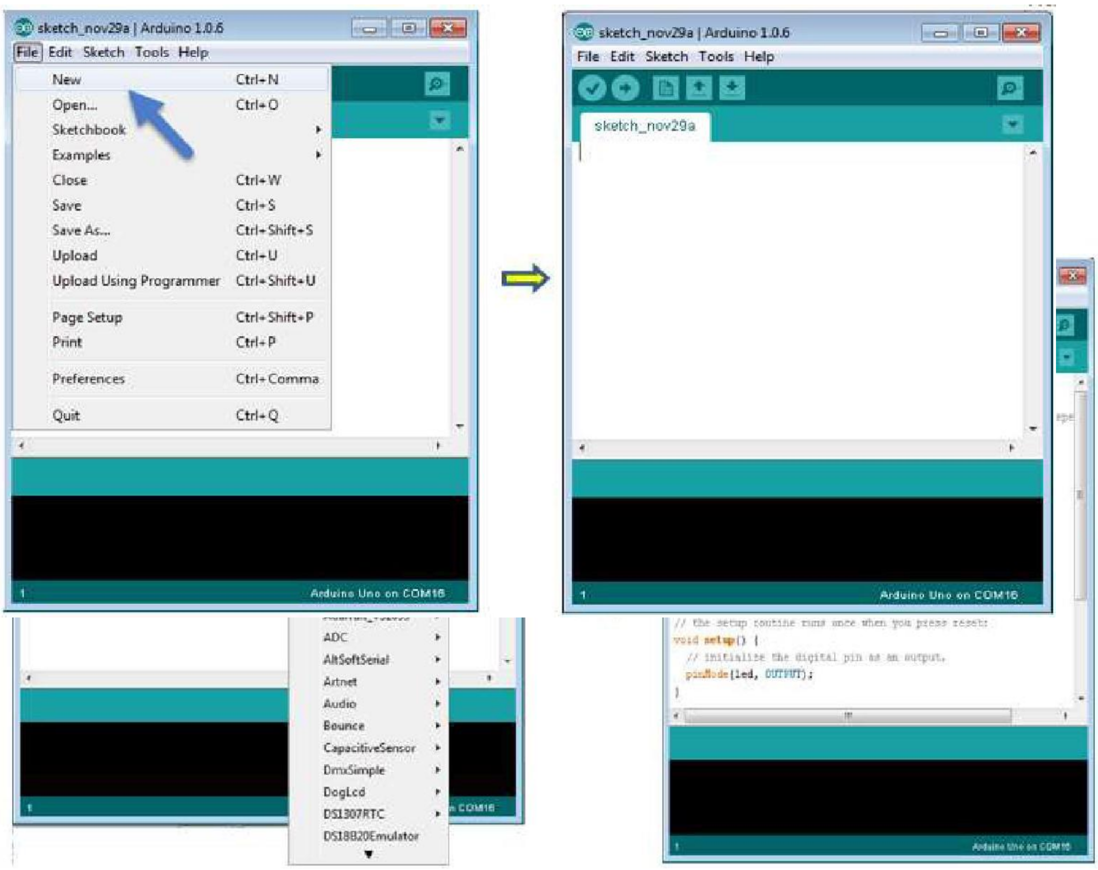


Step 5: Open your first project.

Once the software starts, you have two options:

- Create a new project.
- Open an existing project example.

To create a new project, select File --> New. To open

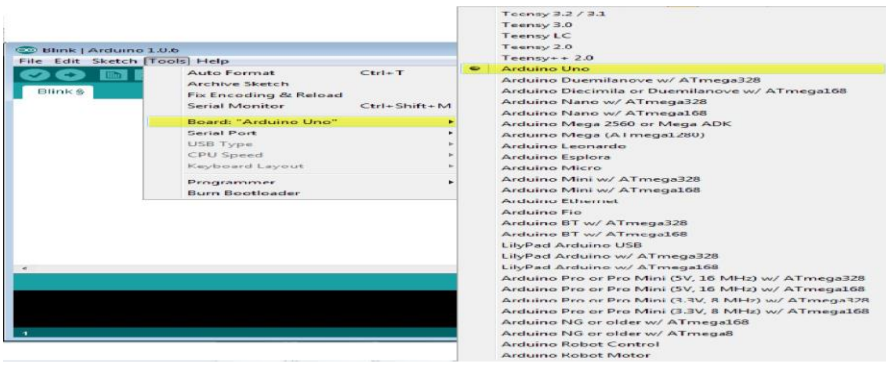


To open an existing project example, select File -> Example -> Basics -> Blink.

Here, we are selecting just one of the examples with the name **Blink**. It turns the LED on and off with sometime delay. You can select any other example from the list.

Step 6: Select your Arduino board.

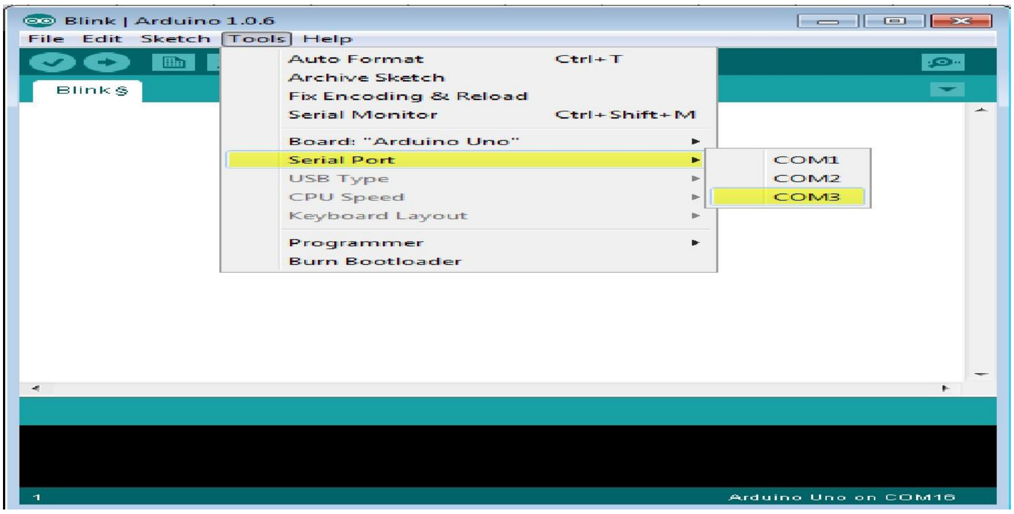
To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer.



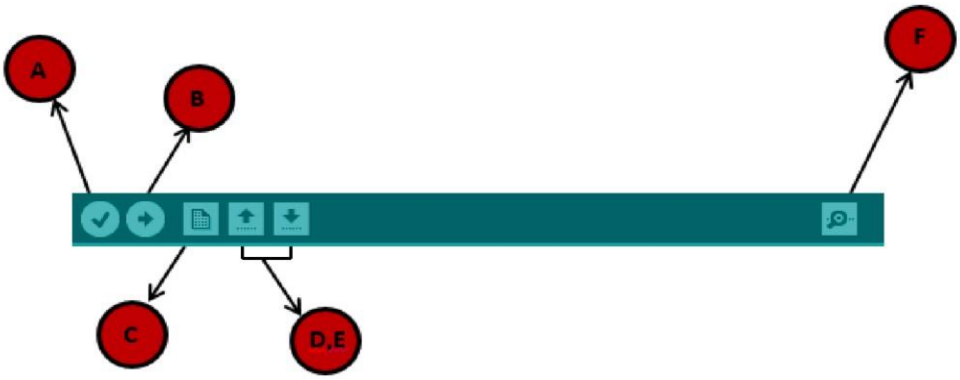
Go to Tools -> Board and select your board

Here, we have selected Arduino Uno board according to our tutorial, but you must select the name matching the board that you are

Step 7: Select your serial port. Select the serial device of the Arduino board. Go to **Tools** ->**Serial Port** menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serial port.



Step 8: Upload the program to your board. Before explaining how we can upload our program to the board, we must demonstrate the function of each symbol appearing in the Arduino IDE toolbar.



A- Used to check if there is any compilation error.

B- Used to upload a program to the Arduino board.

C- Shortcut used to create a new sketch.

D- Used to directly open one of the example sketch.

E- Used to save your sketch.

F- Serial monitor used to receive serial data from the board and send the serial data to board.

Now, simply click the "Upload" button in the environment. Wait a few seconds; you will see the RX and TX LEDs on the board, flashing. If the upload is successful, the message "Done uploading" will appear in the status bar.

Note: If you have an Arduino Mini, NG, or other board, you need to press the reset button physically on the board, immediately before clicking the upload button on the Arduino Software.

Arduino programming structure

In this chapter, we will study in depth, the Arduino program structure and we will learn more new terminologies used in the Arduino world. The Arduino software is open-source. The source code for the Java environment is released under the GPL and the C/C++ microcontroller libraries are under the LGPL.

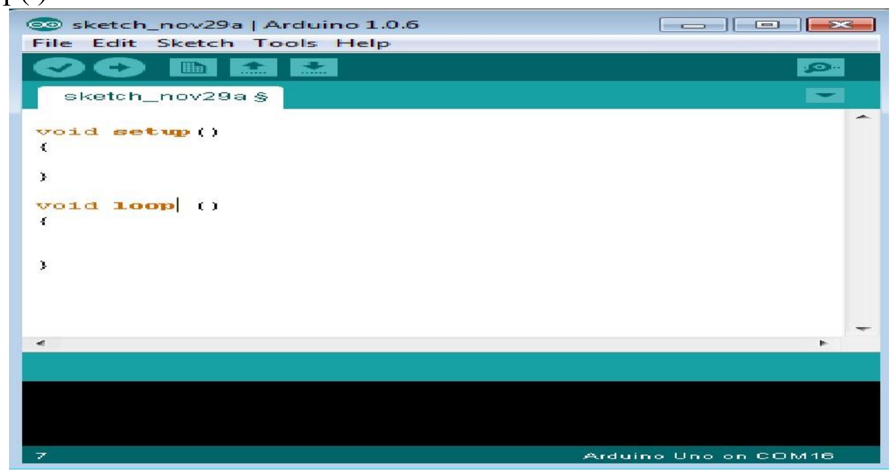
Sketch: The first new terminology is the Arduino program called "**sketch**".

Structure

Arduino programs can be divided in three main parts: **Structure**, **Values** (variables and constants), and **Functions**. In this tutorial, we will learn about the Arduino software program, step by step, and how we can write the program without any syntax or compilation error.

Let us start with the **Structure**. Software structure consist of two main functions:

- Setup( ) function
- Loop( ) function



```
Void setup ( )
    {
    }
```

PURPOSE:

The **setup ()** function is called when a sketch starts. Use it to initialize the variables, pin modes, start using libraries, etc. The setup function will only run once, after each power up or reset of the Arduino board.

INPUT

OUTPUT

RETURN

```
Void Loop ()
    {
    }
```

PURPOSE:

After creating a **setup ()** function, which initializes and sets the initial values, the **loop ()** function does precisely what its name suggests, and loops secutively, allowing your program to change and respond. Use it to actively control the Arduino board.

CHAPTER 6

RESULT

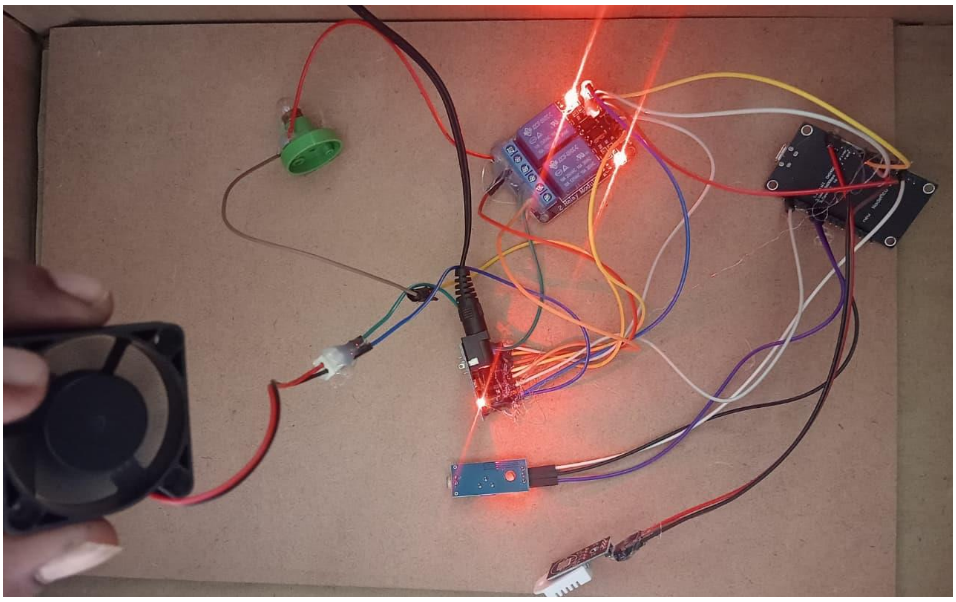


Fig 6.1: Experimental Setup

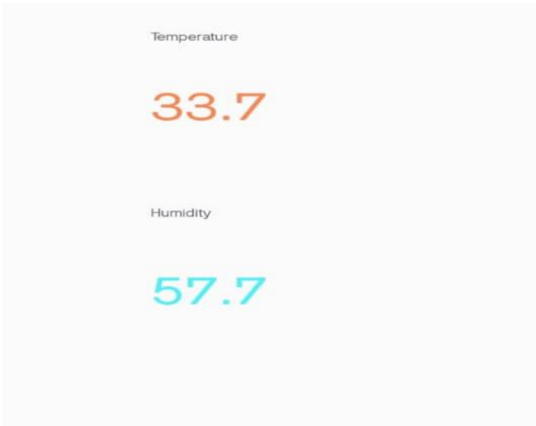


Fig 6.2: Output

- The smart incubator was tested for egg hatching under controlled temperature, humidity, and turning conditions.
- Final results from smart incubators for chicken farming show significantly improved hatch rates, operational efficiency, and remote monitoring capabilities compared to traditional methods. By precisely controlling critical environmental factors, these incubators increase productivity and profitability for both small and large-scale farmers.

CHAPTER 7

ADVANTAGES & APPLICATIONS

7.1 ADVANTAGES

- **Automated Temperature Control** – Maintains optimal warmth for eggs.
- **Humidity Regulation –** Keeps eggs from drying or getting too moist.
- **Automatic Egg Turning –** Prevents embryo from sticking to the shell.
- **Higher Hatchability Rate –** Stable conditions lead to more healthy chicks.
- **Labor Saving –** Reduces need for constant manual supervision.
- **Time Efficiency –** Monitoring and adjustments are automated.
- **Remote Monitoring –** Alerts via mobile apps or sensors.
- **Energy Efficient –** Optimized use of power for heating and fans.
- **Scalable Capacity –** Can handle small or large batches of eggs.
- **Consistent Quality Chicks –** Uniform growth and lower mortality.
- **Better Biosecurity –** Reduced human handling lowers disease risk.
- **Data Recording –** Tracks temperature, humidity, and hatch rates for analysis.

7.2 APPLICATIONS

- **Commercial Hatcheries** – Large-scale chick production.
- **Small Poultry Farms** – Backyard or small-scale incubation.
- **Research & Education** – Studying embryology and poultry development.
- **Improving Hatchability** – Ensures uniform growth and healthy chicks.
- **Remote Monitoring Farms** – Farms using IoT for smart management.
- **Controlled Breeding Programs** – Selective breeding of high-quality poultry.
- **Biosecure Hatching** – Reduces risk of disease from manual handling.
- **Training Purposes** – Teaching modern poultry farming techniques.
- **On-Farm Hatching Systems** – Immediate access to feed and water after hatching.
- **Quality Assurance** – Uniform growth and stronger chicks.
- **Reducing Labor Costs** – Automated monitoring and egg handling.

CHAPTER 8

CONCLUSION & FUTER SCOPE

8.1 CONCLUSION

The Smart Incubator for Poultry Farming (Chicks) is a modern and technology-driven system designed to create a safe, controlled, and healthy environment for chicks after hatching. It automatically manages essential factors like temperature, humidity, air circulation, and light intensity, ensuring that chicks grow comfortably without stress or environmental imbalance.

This system uses sensors, microcontrollers (like NodeMCU), and IoT technology to continuously monitor the living conditions of chicks. Data collected from these sensors helps to maintain the ideal environment, reducing manual supervision and human error. Through real-time updates, farmers can monitor and control the system remotely, improving efficiency and convenience.

The smart incubator contributes significantly to reducing chick mortality, improving growth rates, and ensuring consistent health and development. It also minimizes labor costs and operational difficulties in traditional chick-rearing methods. In the future, the integration of AI and data analytics could make the system even more intelligent—capable of predicting health issues and adjusting conditions automatically.

Overall, the Smart Incubator for Chicks represents a major step toward smart, automated, and sustainable poultry farming, improving both productivity and animal welfare while empowering farmers with modern agricultural technology.

Hence, the Smart Incubator for Poultry Farming (Chicks) is not only an innovation but a practical and futuristic solution for improving productivity, reducing losses, and ensuring better animal welfare in the poultry industry.

- **Maintenance Requirement** – Sensors and electronic components need regular calibration and maintenance to ensure accuracy and reliability.
- **Technical Knowledge Needed** – Farmers must have basic technical skills to operate, troubleshoot, and manage the smart incubator effectively.
- **Internet Connectivity Issues** – IoT-based monitoring requires stable internet; rural areas with poor connectivity may face difficulties.

8.2 FUTURE SCOPE

The future scope of the Smart Incubator for Poultry Farming (Chicks) is vast and promising as technology continues to advance in the field of smart agriculture. In the coming years, this system can be further enhanced with artificial intelligence (AI), machine learning (ML), and data analytics to predict and maintain the ideal environment for chick growth automatically. These technologies will enable the incubator to learn from real-time data, adapt to different climatic conditions, and ensure better health and growth of chicks.

Future models can include automated feeding and watering systems, which will further reduce manual effort and ensure that chicks receive consistent nutrition and hydration. Additionally, health monitoring sensors can be integrated to detect early signs of disease or stress among chicks, allowing farmers to take preventive action in time.

- Advanced Temperature Control – Future systems can use high-precision temperature sensors to maintain stable and uniform warmth for chicks.
- Automatic Humidity Regulation – Intelligent humidity sensors can automatically adjust moisture levels for a comfortable environment.
- Smart Fan Speed Control – Fans can be programmed to adjust their speed based on real-time temperature and humidity changes.
- Energy-Efficient Bulb System – LED or infrared bulbs can provide controlled heating and lighting while saving energy.
- AI-Based Environment Adjustment – AI can automatically balance temperature and humidity by controlling fans and bulbs efficiently.
- Real-Time Monitoring – Temperature and humidity data can be tracked continuously using IoT for precise control.
- Automatic Alerts – The system can send alerts when temperature or humidity goes beyond safe levels.
- Solar-Powered Heating and Cooling – Future incubators can use solar energy to power bulbs and fans for sustainable operation.
- Uniform Air Circulation – Improved fan design will ensure even airflow and prevent hot or cold spots.
- Smart Lighting Control – Bulbs can automatically adjust brightness to simulate natural day and night cycles for chicks.

REFERENCES

1. **Karim, M. Hasnat, Navid Newaz, and Md. Ijtihad Abtahi (2025).** This paper presents an IoT-based smart chicken brooder designed to automate environmental and resource management for chicks. The system uses an ATmega2560 microcontroller and DHT22 sensors, with remote monitoring and SMS notifications.

2. **Jagatheeshbabu, S., S.T. Selvan, and G. Sujatha (2025).** These authors detail the "Design and Development of SMART Brooder Using Raspberry Pi," focusing on energy conservation and enhancing poultry welfare. Their work uses a Raspberry Pi Pico W for real-time monitoring and control.

3. **Mahmud, Imon, et al. (2025).** In their work on an "IoT-Based Smart Poultry and Hatchery Farm," the authors cover integrated solutions for efficient farming, including smart brooding.

4. **Godinho, António, et al. (2025).** This conceptual study on "Wireless Environmental Monitoring and Control in Poultry Houses" discusses using IoT to maintain optimal conditions for brooding, focusing on environmental control.

5. **Muhammad, Faiz Haji Hambali, Ravi Kumar Patchmuthu, and Au Thien Wan (2025).** Their research, aimed at reducing chick mortality, developed an automated IoT-based brooder that monitors temperature, humidity, air quality, and feeding.

6. **Muttha, Rupesh I. et al. (2014).** "PLC Based Poultry Automation System". This paper explores the use of Programmable Logic Controllers (PLCs) in automating poultry farms. It demonstrates an early implementation of automation technology for managing the environmental conditions of chicks.

7. **Tibot Technologies (2021) / T-Moov (2022)Citation.** Commercial autonomous mobile robots, T-Moov and Spoutnic NAV.Cited in research on robotics for poultry farming, these commercial robots are designed to move autonomously within a poultry house. Their motion triggers chicks to be more active, leading to improved health and reduced issues like floor eggs in free-range systems.

APPENDIX

```
/**************************************************
 * NodeMCU + DHT22 + Blynk Cloud + LED + FAN + BULB Control
 *
 * DHT22 â†’ D3 (GPIO 0)
 * LED   â†’ D4
 * FAN   â†’ D5
 * BULB  â†’ D6
 * WiFi: SSID: rpi02 | Password: raspberry
 * Blynk Template ID: TMPL330NjWzPV
 * Template Name: NodeMCU
 * Auth Token: WOLIpvA7_2SyfGM-hZ4a-ZViYHR7gC9z
 **************************************************/
#define BLYNK_TEMPLATE_ID "TMPL3XieEvxVn"
#define BLYNK_TEMPLATE_NAME "Smart Incubation for Poultry Farming"
#define BLYNK_AUTH_TOKEN "ahHm6QaBXBJNu2IzpvI8NKwj73FffXjy"
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include "DHT.h"
// WiFi credentials
char ssid[] = "batch09";
char pass[] = "batch0909";
// DHT configuration
#define DHTPIN D4
```

```
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);
// LED, FAN & BULB configuration
#define LED_PIN D3     // LED pin
#define FAN_PIN D5     // FAN pin
#define BULB_PIN D6    // BULB pin
BlynkTimer timer;
/ Function to read DHT22 and send data to Blynk, control FAN & BULB
void sendSensorData() {
  float h = dht.readHumidity();
  float t = dht.readTemperature();
 if (isnan(h) || isnan(t)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
  }
  float hic = dht.computeHeatIndex(t, h, false);
  Serial.print("Humidity: ");
  Serial.print(h);
  Serial.print("%  Temperature: ");
  Serial.print(t);
  Serial.print("Â°C  Heat Index: ");
  Serial.print(hic);
  Serial.println("Â°C");
  // Send sensor data to Blynk app
```

```
  Blynk.virtualWrite(V0, t);   // Temperature
  Blynk.virtualWrite(V1, h);   // Humidity
  // Fan control logic: turn ON fan if temp > 30Â°C
  if (t > 31.0) {
    digitalWrite(FAN_PIN, LOW); // Turn ON fan (active LOW relay)
    Serial.println("Fan: ON (Temp > 30Â°C)");
    Blynk.virtualWrite(V3, 1); // Optionally update a Blynk widget for fan status
  } else {
    digitalWrite(FAN_PIN, HIGH);  // Turn OFF fan
    Serial.println("Fan: OFF (Temp <= 30Â°C)");
    Blynk.virtualWrite(V3, 0); // Optionally update a Blynk widget for fan status
  }
  // Bulb control logic: turn ON bulb if temp < 27Â°C
  if (t < 30.5) {
    digitalWrite(BULB_PIN, LOW); // Turn ON bulb (active LOW relay)
    Serial.println("Bulb: ON (Temp < 27Â°C)");
    Blynk.virtualWrite(V4, 1); // Optionally update a Blynk widget for bulb status
  } else {
    digitalWrite(BULB_PIN, HIGH); // Turn OFF bulb
    Serial.println("Bulb: OFF (Temp >= 27Â°C)");
    Blynk.virtualWrite(V4, 0); // Optionally update a Blynk widget for bulb status
  }
}
// Function to handle LED control from Blynk app
```

```
 BLYNK_WRITE(V2) {  // V2 is the virtual pin for LED control
  int ledState = param.asInt(); // 1 = ON, 0 = OFF
  digitalWrite(LED_PIN, ledState);
  Serial.print("LED State: ");
  Serial.println(ledState ? "ON" : "OFF");
 } void setup() {
   Serial.begin(9600);
  Serial.println("Connecting to Blynk...");
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, HIGH); // Turn off LED initially
  pinMode(FAN_PIN, OUTPUT);
  digitalWrite(FAN_PIN, HIGH); // Turn off FAN initially (Active LOW relay)
  pinMode(BULB_PIN, OUTPUT);
  digitalWrite(BULB_PIN, HIGH); // Turn off BULB initially (Active LOW relay)
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  dht.begin();
  timer.setInterval(2000L, sendSensorData); // Send DHT22 data every 2 seconds
}
void loop() {
  Blynk.run();
  timer.run();
}
```